

# **MATH I – Week 1**

Fall 2026

**Lecture presentation Part C**

on

**Complex numbers**

# Complex numbers

## 1. Why complex numbers are needed

Some algebraic equations have no solutions in the real numbers. For example,

$$x^2 + 1 = 0$$

requires  $x^2 = -1$ , but no real number has a negative square. We therefore introduce the **imaginary unit**

$$i = \sqrt{-1}, \quad i^2 = -1.$$

A **complex number** has the form

$$z = a + bi,$$

where  $a, b \in \mathbb{R}$ . The number  $a$  is the **real part**, and  $b$  is the **imaginary part**:

$$\operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b.$$

For example, if  $z = 4 - 3i$ , then

$$\operatorname{Re}(z) = 4, \quad \operatorname{Im}(z) = -3.$$

The set of complex numbers is denoted by

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

Every real number is also complex because

$$a = a + 0i.$$

Thus,

$$\mathbb{R} \subseteq \mathbb{C}.$$

# Equality of complex numbers

## 2. Equality of complex numbers ¶

Two complex numbers are equal if their real parts are equal and their imaginary parts are equal:

$$a + bi = c + di$$

if and only if

$$a = c \quad \text{and} \quad b = d.$$

For example, if

$$(2x - 1) + (y + 3)i = 5 - 2i,$$

then

$$2x - 1 = 5, \quad y + 3 = -2.$$

Therefore,

$$x = 3, \quad y = -5.$$

# Powers of $i$

## 3. Powers of $i$

The powers of  $i$  repeat every four exponents:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i,$$

and

$$i^4 = 1.$$

The cycle then repeats:

$$i^{n+4} = i^n.$$

To calculate a large power of  $i$ , divide the exponent by 4 and use the remainder.

# Operations

## 4. Algebraic operations

Let

$$z_1 = a + bi, \quad z_2 = c + di.$$

### 4.1 Addition

Add the real parts and imaginary parts separately:

$$z_1 + z_2 = (a + c) + (b + d)i.$$

Example:

$$(3 + 2i) + (5 - 7i) = 8 - 5i.$$

### 4.2 Subtraction

$$z_1 - z_2 = (a - c) + (b - d)i.$$

Example:

$$(3 + 2i) - (5 - 7i) = -2 + 9i.$$

# Algebraic operations (cont'd)

## 4.3 Multiplication

Use the distributive law and  $i^2 = -1$ :

$$(a + bi)(c + di) = ac + adi + bci + bdi^2.$$

Therefore,

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Example:

$$(2 + 3i)(4 - i) = 8 - 2i + 12i - 3i^2 = 11 + 10i.$$

# Algebraic operations (cont'd)

## 4.4 Division

To divide complex numbers, multiply the numerator and denominator by the conjugate of the denominator.

For example,

$$\frac{3 + 2i}{1 - i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)}.$$

The denominator is

$$(1 - i)(1 + i) = 1 - i^2 = 2.$$

The numerator is

$$(3 + 2i)(1 + i) = 3 + 3i + 2i + 2i^2 = 1 + 5i.$$

Hence,

$$\frac{3 + 2i}{1 - i} = \boxed{\frac{1}{2} + \frac{5}{2}i}.$$

# The complex conjugate of $z$

## 5. Complex conjugate

The **complex conjugate** of

$$z = a + bi$$

is

$$\boxed{\bar{z} = a - bi}.$$

Geometrically, conjugation reflects a complex number across the real axis.

Important properties include

$$z + \bar{z} = 2 \operatorname{Re}(z),$$

$$z - \bar{z} = 2i \operatorname{Im}(z),$$

and

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2.$$

# 6. The complex plane

A complex number

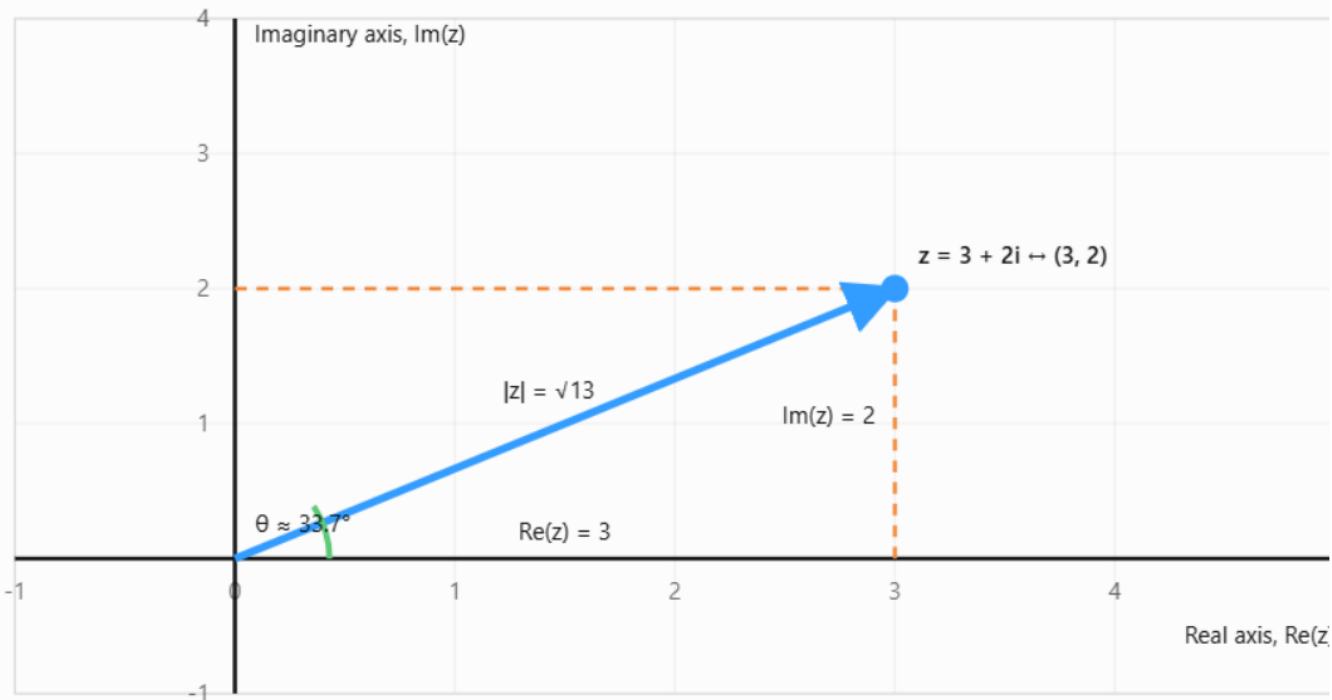
## The complex plane

$$z = a + bi$$

can be represented by the point  $(a, b)$  in the **complex plane**, also called the **Argand plane**.

- The horizontal axis is the real axis.
- The vertical axis is the imaginary axis.

Complex number on the complex plane:  $z = 3 + 2i$



Thus,

$$\text{Re}(z) = 3, \quad \text{Im}(z) = 2.$$

The distance from the origin is the **modulus**:

$$|z| = \sqrt{3^2 + 2^2} = \sqrt{13},$$

and the angle with the positive real axis is the **argument**:

$$\arg(z) = \tan^{-1}\left(\frac{2}{3}\right) \approx 33.7^\circ.$$

For a nonzero complex number  $z = a + bi$ , the **argument** of  $z$  is an angle  $\theta$  measured from the positive real axis to the vector representing  $z$ .

We write

$$\theta = \arg z.$$

Since adding a full rotation does not change the direction,

$$\arg z = \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

A selected value, usually in

$$(-\pi, \pi],$$

is called the **principal argument** and is denoted by

$$\text{Arg}(z).$$

Although

$$\tan \theta = \frac{b}{a},$$

the quadrant containing  $(a, b)$  must be considered.

## In-Class activity

Exercise:

In which quadrant

$$z = -1 + i$$

lies?

# How else can we express Complex Numbers?

## 9. Algebraic, trigonometric and exponential forms

### 9.1 Algebraic form

The usual form is

$$z = a + bi.$$

### 9.2 Trigonometric form ¶

If  $r = |z|$  and  $\theta = \arg z$ , then

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Therefore,

$$z = r(\cos \theta + i \sin \theta).$$

This is the **trigonometric** or **polar form**.

For example, consider

$$z = 1 + \sqrt{3}i.$$

Its modulus is

$$r = \sqrt{1^2 + (\sqrt{3})^2} = 2.$$

Since the point is in the first quadrant,

$$\theta = \frac{\pi}{3}.$$

Thus,

$$1 + \sqrt{3}i = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right).$$

cont'd

### 9.3 Euler's formula

Euler's formula states that

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

It gives the famous identity

$$e^{i\pi} + 1 = 0.$$

### 9.4 Exponential form

Using Euler's formula, the trigonometric form becomes

$$z = re^{i\theta}.$$

Thus,

$$1 + \sqrt{3}i = 2e^{i\pi/3}.$$

Because the argument is not unique, all equivalent exponential forms are

$$z = re^{i(\theta+2k\pi)}, \quad k \in \mathbb{Z}.$$

# de Moivre's formula

## 11. de Moivre's formula

For every integer  $n$ ,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Equivalently, if

$$z = r(\cos \theta + i \sin \theta),$$

then

$$z^n = r^n(\cos(n\theta) + i \sin(n\theta)).$$

In exponential form,

$$(re^{i\theta})^n = r^n e^{in\theta}.$$

**Example: Calculate  $(1 + i)^8$**

First write  $1 + i$  in polar form:

$$|1 + i| = \sqrt{2}, \quad \arg(1 + i) = \frac{\pi}{4}.$$

Therefore,

$$1 + i = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

By de Moivre's formula,

$$(1 + i)^8 = (\sqrt{2})^8 \left( \cos \frac{8\pi}{4} + i \sin \frac{8\pi}{4} \right).$$

Since

$$(\sqrt{2})^8 = 16, \quad \cos(2\pi) = 1, \quad \sin(2\pi) = 0$$

we get

$$(1 + i)^8 = 16.$$

# 12. Roots of complex numbers

Suppose we want to solve

$$z^n = w,$$

where

$$w = re^{i\theta}.$$

Write

$$z = \rho e^{i\phi}.$$

Then

$$z^n = \rho^n e^{in\phi}.$$

Therefore,

$$\rho^n = r \implies \rho = r^{1/n},$$

and

$$n\phi = \theta + 2k\pi.$$

Hence the  $n$  distinct roots are

$$z_k = r^{1/n} \left[ \cos \left( \frac{\theta + 2k\pi}{n} \right) + i \sin \left( \frac{\theta + 2k\pi}{n} \right) \right],$$

where

$$k = 0, 1, \dots, n - 1.$$

The roots have equal modulus and are equally spaced around a circle.

# In-Class activity

## Exercise 1: Basic complex arithmetic

Given

$$z_1 = 2 + 3i, \quad z_2 = 1 - 5i,$$

calculate:

$$z_1 + z_2, \quad z_1 - z_2, \quad z_1 z_2, \quad \frac{z_1}{z_2}.$$

Write every answer in the form  $a + bi$ .

---

## Exercise 2: Explain why

$$1 + i\sqrt{3} = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2e^{i\frac{\pi}{3}}$$

---

## Exercise 3: Calculate $(1 + i)^6$

# Solutions

## Exercise 1: Basic complex arithmetic

Given

$$z_1 = 2 + 3i, \quad z_2 = 1 - 5i,$$

and using  $i^2 = -1$ :

**Addition**

$$z_1 + z_2 = (2 + 3i) + (1 - 5i)$$

$$\boxed{z_1 + z_2 = 3 - 2i}.$$

**Subtraction**

$$z_1 - z_2 = (2 + 3i) - (1 - 5i)$$

$$\boxed{z_1 - z_2 = 1 + 8i}.$$

**Multiplication**

$$\begin{aligned}z_1 z_2 &= (2 + 3i)(1 - 5i) \\&= 2 - 10i + 3i - 15i^2 \\&= 2 - 7i + 15.\end{aligned}$$

Therefore,

$$\boxed{z_1 z_2 = 17 - 7i}.$$

**Division**

Multiply the numerator and denominator by the conjugate of  $1 - 5i$ , which is  $1 + 5i$ :

$$\frac{z_1}{z_2} = \frac{2 + 3i}{1 - 5i} \cdot \frac{1 + 5i}{1 + 5i}.$$

The numerator is

$$(2 + 3i)(1 + 5i) = 2 + 10i + 3i + 15i^2 = -13 + 13i.$$

The denominator is

$$(1 - 5i)(1 + 5i) = 1 - (5i)^2 = 26.$$

Thus,

$$\boxed{\frac{z_1}{z_2} = -\frac{1}{2} + \frac{1}{2}i}.$$

## Exercise 2: Trigonometric and exponential forms

For

$$z = 1 + i\sqrt{3},$$

the real and imaginary parts are

$$a = 1, \quad b = \sqrt{3}.$$

The modulus is

$$r = |z| = \sqrt{a^2 + b^2} = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2.$$

Because  $z$  lies in the first quadrant, its argument is

$$\theta = \arctan\left(\frac{b}{a}\right) = \arctan(\sqrt{3}) = \frac{\pi}{3}.$$

Therefore, its trigonometric form is

$$z = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right).$$

Indeed,

$$2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2 \left( \frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = 1 + i\sqrt{3}.$$

Using Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

we also obtain

$$2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2e^{i\pi/3}.$$

Hence,

$$\boxed{1 + i\sqrt{3} = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2e^{i\pi/3}}.$$

### Exercise 3: Calculate $(1 + i)^6$

Using direct algebra,

$$(1 + i)^2 = 1 + 2i + i^2 = 2i.$$

Therefore,

$$(1 + i)^6 = ((1 + i)^2)^3 = (2i)^3.$$

Since  $i^3 = -i$ ,

$$(2i)^3 = 8i^3 = -8i.$$

Thus,

$$\boxed{(1 + i)^6 = -8i}.$$

Alternatively, since

$$1 + i = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right),$$

de Moivre's formula gives

$$\begin{aligned} (1 + i)^6 &= (\sqrt{2})^6 \left( \cos \frac{6\pi}{4} + i \sin \frac{6\pi}{4} \right) \\ &= 8 \left( \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) \\ &= 8(0 - i) \\ &= \boxed{-8i}. \end{aligned}$$